Reviewer Pack — Secant Variety Dimension Lower Bounds Disjointness Communica…

Entry 059c72c72c3c · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Secant Variety Dimension Lower Bounds Disjointness Communication Complexity

Entry ID: `059c72c72c3c` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-10 04:55:12 UTC

1. Conjecture

Field A (mathematical branch): Secant Variety Geometry **Field B** (complexity object): Communication Complexity of DISJOINTNESS

Statement:

> For a random n×n DISJOINTNESS matrix M, the dimension of its secant variety satisfies dim(σ(M)) ≥ Ω(n). This holds for all instances where the communication matrix has no zero rows or columns.

Rationale (proposer's reasoning):

> Secant variety dimensions capture the algebraic structure of communication matrices. For DISJOINTNESS, the high-dimensional geometry of its matrix (which is a 0-1 matrix with row/column support constraints) may enforce a lower bound on the secant variety's dimension, mirroring the Ω(n) communication complexity lower bound.

Taxonomy category: `COMM_DISJ` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `3179a19f74707a18`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM_1	LLM_2
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def gaussian_elimination(A):
    n = len(A)
    for i in range(n):
        # Find pivot in column i
        max_row = i
        for j in range(i+1, n):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]

        # Eliminate entries below pivot
        factor = A[i][i]
        for j in range(i+1, n):
            A[j][i] /= factor

    rank = 0
    for row in A:
        if any(row):
            rank += 1
    return rank

def matrix_rank(M):
    A = [row[:] for row in M]
    return gaussian_elimination(A)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 10
    M = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]

    # Ensure no zero rows or columns
    while any(sum(row) == 0 for row in M):
        M = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
    while any(sum(col) == 0 for col in zip(*M)):
        M = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
```

```
    dim_secant_variety = matrix_rank(M)
    metric_value = dim_secant_variety
    instances_tested = 1

    conjecture_holds = dim_secant_variety >= 0.8 * n
    counterexample = "" if conjecture_holds else "dim(σ(M)) < 0.8n"

    return {
        "metric_name": "secant_variety_dimension",
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(result
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample='dim(σ(M)) < 0.8n' first_failing_seed={first_fail
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 54, in run_trial
    dim_secant_variety = matrix_rank(M)
                         ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 41, in matrix_rank
    return gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 31, in gaussian_elim
```

```
    A[j][i] /= factor
ZeroDivisionError: division by zero
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity. | next: Fix numerical stability in Gaussian elimination to handle singular matrices

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	111924
2	propose	ollama_remote	qwen3:8b	0	0	84785
3	propose	ollama_remote	qwen3:8b	0	0	111765
4	propose	ollama_remote	qwen3:8b	0	0	68964
5	preregistration	ollama_remote	qwen3:8b	0	0	24074
6	novelty	ollama_remote	qwen3:8b	0	0	20654
7	novelty	ollama_remote	qwen3:8b	0	0	16951
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9884
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8725
10	judge	ollama_remote	qwen3:8b	0	0	16572

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 474300 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/059c72c72c3c.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/059c72c72c3c.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/059c72c72c3c.tar.gz` (if generated)